SMART DOOR LOCK SYSTEM

Arduino-Based Smart Access Control and Security System

Student Name	**Samruddhi Talekar**
Roll Number	709
Platform	Tinkercad Simulation

IoT Case Study – Project / Prototype Development

1. Introduction

The Smart Door Lock System is an Arduino-based access-control prototype designed and simulated in Tinkercad. It combines motion detection, distance measurement, password authentication, a display, and a servo motor to demonstrate how a door can be unlocked and locked electronically instead of relying only on a traditional physical key.The system begins in a locked state. When the PIR sensor detects motion, the HC-SR04 ultrasonic sensor measures the distance of the approaching person or object. If the measured distance is less than 100 cm, password entry is enabled. The user enters the password using a keypad. A correct password grants access and moves the servo to the unlocked position. The system then locks again automatically after three seconds.

1.1 Problem Statement

Traditional door locks depend on physical keys, which can be lost, misplaced, or duplicated. They also provide limited electronic feedback about access attempts. This project demonstrates a prototype that uses sensors and password-based authentication to control access and provide visual and audio status feedback.

1.2 Objectives

- Detect nearby motion using a PIR sensor.
- Measure the distance of the detected object using an HC-SR04 ultrasonic sensor.
- Enable password entry only when the measured distance is below 100 cm.
- Use a keypad to authenticate access.
- Display system status on an LCD.
- Control the locking mechanism using a servo motor.
- Provide audio feedback through a buzzer for successful and unsuccessful attempts.
- Automatically return the lock to the locked position after three seconds.

2. Components Used

Component	Purpose
Arduino Uno	Main controller that reads sensor inputs and controls the system outputs.
PIR motion sensor	Detects movement in the surrounding area.
HC-SR04 ultrasonic sensor	Measures distance to an approaching object.
4 × 4 keypad	Allows the user to enter the access password.
LCD display (I2C)	Displays status and access-related messages.
Servo motor	Represents the door-lock mechanism by changing its angle.
Buzzer	Provides sound feedback for successful or unsuccessful password entry.
Breadboard and jumper wires	Provide connections between the components.

3. Design and Circuit Screenshot

The circuit was assembled and tested in Tinkercad using an Arduino Uno, PIR sensor, HC-SR04 ultrasonic sensor, keypad, LCD, servo motor, and buzzer. The components are connected with jumper wires, with the Arduino coordinating the sensor readings and the locking response.

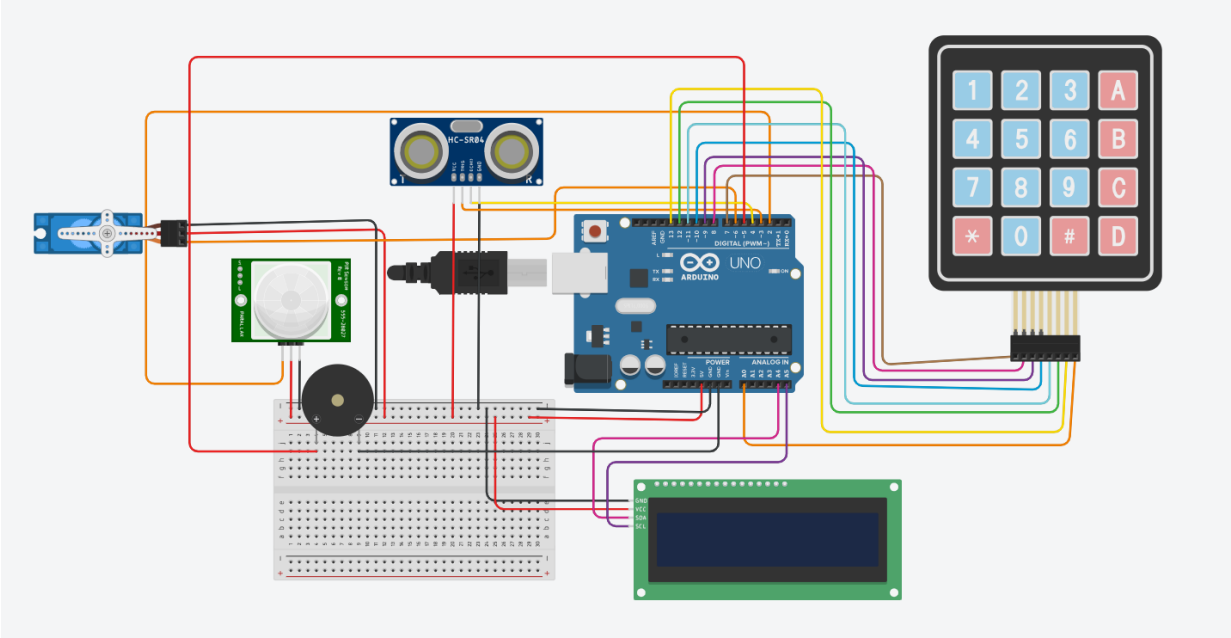


Figure 1. Tinkercad circuit design of the Smart Door Lock System.

4. Working Principle

1. **Initial state:** When the system starts and no motion is detected, the door remains locked. The servo is at 0°.
2. **Motion detection:** The PIR sensor detects movement near the door.
3. **Distance verification:** The HC-SR04 ultrasonic sensor measures the distance of the detected person or object.
4. **Distance condition:** If the measured distance is less than 100 cm, password entry is enabled.
5. **Password entry:** The user enters the password through the keypad. The password shown in the presentation is 1234#.
6. **Correct password:** The system grants access, gives a short buzzer beep, and moves the servo to 90° to represent unlocking.
7. **Automatic relocking:** After three seconds, the servo returns to 0° and the door locks automatically.
8. **Incorrect password:** The system displays a wrong-password/try-again message, produces a long buzzer sound, and keeps the servo in the locked position.

5. Code

The Arduino program coordinates the PIR sensor, ultrasonic distance sensor, keypad, LCD, servo motor, and buzzer. Its logic follows the sequence described in the working-principle section: keep the lock closed by default, detect motion, check distance, accept password input when the distance condition is satisfied, and provide the appropriate response.The PowerPoint supplied for this project describes the program's behaviour but does not include the complete Arduino source code. For that reason, the exact code is not reproduced in this report. The final report can include the original Tinkercad/Arduino code in this section if it is available separately.

6. Real-World Applications and Existing Product

Smart door locks are used in homes, apartments, offices, and other places where controlled access is required. They can replace or supplement physical keys with PIN codes, electronic credentials, automatic locking, and access feedback.

Example product: Yale YDR 414A smart lock. Yale describes this product as supporting access methods such as fingerprint, PIN code, remote control, and mechanical-key override, with optional connected features through the Yale Home ecosystem. Source: Yale Home – YDR 414A (https://www.yalehome.com/in/en/products/smart-products/smart-locks/connected/ydr414-rim-lock).

6.1 What This Prototype Adds / Demonstrates

- Motion detection with a PIR sensor before access is attempted.
- Distance-based gating: keypad access is enabled only when the measured distance is below 100 cm.
- Password authentication through a keypad.
- LCD status messages and different buzzer feedback for correct and incorrect passwords.
- Servo-based unlocking followed by automatic relocking after three seconds.

Unlike many commercial connected smart locks, this project is a Tinkercad prototype. The supplied project materials do not show mobile-app control, Wi-Fi connectivity, cloud logging, fingerprint access, or installation on a real door; these features should not be considered part of the implemented prototype.

7. Results and Testing

The project presentation reports the following implemented functions:

- Motion detection using the PIR sensor.
- Distance measurement using the HC-SR04 ultrasonic sensor.
- Password authentication through the keypad.
- LCD status display.
- Servo-based locking and unlocking.
- Audio feedback using the buzzer.

The intended test outcomes are: no motion keeps the door locked; motion followed by a distance below 100 cm enables password entry; a correct password unlocks the door briefly; and an incorrect password leaves the door locked while giving an error message and a longer beep.

8. Conclusion

The Smart Door Lock System demonstrates how sensors, embedded programming, password authentication, and automation can be combined to create a basic access-control system. The PIR sensor detects movement, the ultrasonic sensor checks distance, the keypad validates the password, and the servo motor represents the lock mechanism. The LCD and buzzer communicate the system's response to the user. The automatic relocking feature adds an extra layer of convenience by returning the lock to its secure state after three seconds. The Tinkercad simulation provides a practical demonstration of IoT concepts, although a real installation would require suitable hardware, secure password handling, reliable power, and physical lock integration.

References

1. Yale Home, “YDR 414A – RIM Lock.” https://www.yalehome.com/in/en/products/smart-products/smart-locks/connected/ydr414-rim-lock